Practice assignment 10 - Not Graded

This assignment will not be graded and is only for practice.

1 point

Match the functions of two variables in Column A with their directions in which the directional derivatives at (0,0) is maximized given in Column B.

- [ ] (1) $\longrightarrow$ (v)(1) $\longrightarrow$ (v)
- [ ] (1) $\longrightarrow$ (iii)(1) $\longrightarrow$ (iii)
- [ ] (2) $\longrightarrow$ (iv)(2) $\longrightarrow$ (iv)
- [ ] (3) $\longrightarrow$ (iv)(3) $\longrightarrow$ (iv)
- [ ] (2) $\longrightarrow$ (vi)(2) $\longrightarrow$ (vi)
- [ ] (4) $\longrightarrow$ (v)(4) $\longrightarrow$ (v)
- [ ] (3) $\longrightarrow$ (vi)(3) $\longrightarrow$ (vi)
- [ ] (4) $\longrightarrow$ (i)(4) $\longrightarrow$ (i)

No, the answer is incorrect.

Score: 0

Accepted Answers:

(1) $\longrightarrow$ (iii)(1) $\longrightarrow$ (iii)

(2) $\longrightarrow$ (iv)(2) $\longrightarrow$ (iv)

(4) $\longrightarrow$ (v)(4) $\longrightarrow$ (v)

1 point

Which of the following statements is(are) correct?

- [ ] Maximum directional derivative at (0,0) for the function $f(x, y) = |x| + |y|$f(x, y) = |x| + |y| is 1.
- [ ] The graph of the linear approximation to a function $f(x, y)$f(x, y) at a given point PP is the tangent plane to $f(x, y)$f(x, y) at the point PP.
- [ ] A tangent plane to a function $f(x, y)$f(x, y) exists at every point.
- [ ] All directional derivatives of a function $f(x, y)$f(x, y) at a point PP are 0 if the tangent plane is parallel to the XYXY -plane.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The graph of the linear approximation to a function $f(x, y)$f(x, y) at a given point PP is the tangent

plane to $f(x, y)$f(x, y) at the point PP.
All directional derivatives of a function $f(x, y)$f(x, y) at a point PP are 0 if the tangent plane is parallel to the XYXY -plane.
1 point
Which of the following is(are) critical points of the scalar valued multivariable function $f(x, y) = 6x^2 - 2x^3 + 3y^2 - 6xy$f(x, y) = 6x2 − 2x3 + 3y2 − 6xy?

- [ ] (1,1)
- [ ] (0,0)
- [ ] (1,-1)
- [ ] No critical points.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(1,1)
(0,0)

Consider two functions f_1, f_2f1, f2 defined as

$$f_1(x, y, z) = \ln(x^2 + y^2 + 2z^2)\text{f1}(x, y, z) = \ln(x2 + y2 + 2z2)$$

and

$$f_2(x, y, z) = \tan^{-1}(x^2 + y^2 + 2z^2).\text{f2}(x, y, z) = \tan{-1}(x2 + y2 + 2z2).$$

If $L_{f_1}(x, y, z)$Lf1(x, y, z) and $L_{f_2}(x, y, z)$Lf2(x, y, z) are the linear approximation of $f_1(x, y, z)$f1(x, y, z) and $f_2(x, y, z)$f2(x, y, z) at point (1, -1, 1), then find the value of $85(\nabla L_{f_1}(x, y, z) \cdot \nabla L_{f_2}(x, y, z))$85(∇Lf1(x, y, z) · ∇Lf2(x, y, z)).
No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 30.0
1 point

$z = f(x, y) = \frac{-7}{6}x^2 + \frac{1}{3}y^2 + \frac{11}{6}$z = f(x, y) = 6−7x2 + 31y2 + 611 represents a surface. Let (x_0, y_0, z_0)(x0, y0, z0) be the point where the tangent plane to the given surface is parallel to the plane $7x - 8y + 3z + 2 = 0$7x − 8y + 3z + 2 = 0. Find the value of $11x_0 + 5y_0 + 4z_0$11x0 + 5y0 + 4z0.
No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 55
1 point

Find the number of points where the tangent plane to the function $f(x, y) = x^3 + y^3 + 3x^2 - 9x - 12y + 20$f(x, y) = x3 + y3 + 3x2 − 9x − 12y + 20 at the points is parallel to the XYXY -plane.
No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 4
1 point

- Suppose $ax + by + cz = d_1$ ax + by + cz = d1 is a tangent plane to the function $f(x, y) = x^3 + 3y^3 + 9xy$ f(x, y) = x3 + 3y3 + 9xy at (-1,2) and $ax + by + cz = d_2$ ax + by + cz = d2 is a tangent plane to the same function $f(x, y)$ f(x, y) at $(\alpha, 1)$ (α, 1).

  Use the above information to answer questions 7-8:

  1 point

  Which of the following options is(are) true?

  ☐ $\alpha = -2$ α = −2

  ☐ $\alpha = 2$ α = 2

  ☐ There are only two possible candidates for α α.

  ☐ $|d_2 - d_1| = 12$ |d2 − d1| = 12

  No, the answer is incorrect.
  Score: 0
  Accepted Answers:
  $\alpha = 2$ α = 2
  $|d_2 - d_1| = 12$ |d2 − d1| = 12

  Find the value of $\frac{2a + b + c}{d_1 + d_2}$ d1+d2 2a+b+c.

  No, the answer is incorrect.
  Score: 0
  Accepted Answers:
  (Type: Numeric) 1
  1 point

- Let $L(x, y)$ L(x, y) be the linear approximation to $f(x, y) = xe^y + ln(x + y)$ f(x, y) = xey + ln(x + y) at point (1,0).

  Use the above information to answer questions 9-10:

  Suppose $L(x, y) = ax + by + c$ L(x, y) = ax + by + c, where $a, b, c \in R$ a, b, c ∈ R. Find the value of $a + b - c$ a + b − c.

  No, the answer is incorrect.
  Score: 0
  Accepted Answers:
  (Type: Numeric) 5
  1 point

  Find the approximate value of the function $f(x, y)$ f(x, y) at (1.1,0.1) using the linear approximation function $L(x, y)$ L(x, y).

  No, the answer is incorrect.
  Score: 0
  Accepted Answers:
  (Type: Numeric) 1.4
  1 point

  Check Answers

  Your score is: 0/10